

ZANZIBAR, Tanzania, United Republic of

WMO: 638700

Lat: 6.222S

Lon: 39.225E

Elev: 54

StdP: 14.67

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	68.7	70.0	58.4	73.0	76.3	61.8	82.5	79.0	23.4	82.3	22.4	82.4	0.4	130	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.0	93.1	79.6	91.6	78.8	90.3	78.5	81.7	88.8	80.9	87.6	80.2	86.7	11.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	152.3	85.5	79.0	150.9	85.1	78.7	149.0	84.5	45.4	89.7	44.6	88.5	44.0	87.3	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.3	19.1	17.9	DB	63.9	97.5	4.2	3.0	60.8	99.7	58.4	101.5	56.0	103.2	53.0	105.4
			WB	61.4	83.6	3.6	1.3	58.8	84.5	56.7	85.3	54.7	86.1	52.0	87.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	83.7	83.6	83.5	81.8	80.3	79.3	78.5	78.2	78.9	80.1	80.8	82.7	(6)
(7)		DBStd	2.67	1.85	2.10	2.19	2.17	1.81	1.64	1.43	1.61	1.48	1.68	1.66	1.75	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11292	1046	940	1039	955	938	880	882	874	867	934	924	1013	(10)
(11)		CDD65	5817	581	520	574	505	473	430	417	409	417	469	474	548	(11)
(12)		CDH74	55502	6614	5983	6510	5103	4144	3328	2779	2844	3279	4241	4656	6021	(12)
(13)		CDH80	17657	2584	2435	2548	1475	870	648	510	642	921	1345	1483	2196	(13)
(14)	Wind	WSAvg	7.8	8.6	7.5	5.9	7.0	8.6	9.4	10.1	8.3	8.3	7.7	5.5	6.2	(14)
(15)	Precipitation	PrecAvg	61.3	2.8	2.3	7.5	13.1	9.6	2.5	1.7	1.9	2.1	4.0	7.1	6.0	(15)
(16)		PrecMax	85.0	9.8	8.6	16.4	23.7	19.6	8.3	3.2	5.2	6.1	19.3	14.4	11.9	(16)
(17)		PrecMin	32.4	0.0	0.1	2.2	2.0	0.7	0.2	0.6	0.5	0.5	0.2	1.9	0.5	(17)
(18)		PrecStd	12.4	2.2	2.3	3.3	6.3	5.6	2.0	0.9	1.2	1.4	3.9	2.9	3.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	94.9	95.2	91.5	88.1	86.9	86.3	87.1	88.2	90.1	90.2	91.8	(19)
(20)			MCWB	79.3	80.1	80.7	79.7	77.8	75.5	73.7	73.3	73.4	75.5	78.4	78.9	(20)
(21)		2%	DB	91.7	92.9	93.1	89.5	86.7	86.0	85.0	86.0	87.5	89.0	89.3	90.5	(21)
(22)			MCWB	78.3	79.1	80.3	79.5	77.4	75.2	73.4	73.3	73.4	75.6	78.1	78.5	(22)
(23)		5%	DB	91.0	91.3	91.3	88.0	85.9	84.6	84.2	84.6	86.2	87.7	87.9	89.6	(23)
(24)			MCWB	78.1	78.7	79.6	79.4	77.3	74.9	73.2	73.0	73.4	75.8	78.2	78.5	(24)
(25)		10%	DB	89.5	89.7	89.6	86.7	84.5	83.9	82.8	83.9	85.1	86.2	86.4	88.2	(25)
(26)			MCWB	78.0	78.3	79.1	79.1	77.0	74.8	73.1	72.9	73.5	75.8	78.0	78.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	81.9	83.4	82.3	80.1	78.4	76.8	76.5	77.4	79.6	81.0	81.9	(27)
(28)			MCD B	89.8	90.6	92.2	88.3	85.0	83.5	82.8	83.0	83.5	85.3	86.4	88.3	(28)
(29)		2%	WB	80.6	80.7	81.7	80.9	79.3	77.4	75.6	75.5	76.4	78.6	80.1	80.7	(29)
(30)			MCD B	88.1	89.2	89.4	86.6	84.3	82.7	81.5	81.9	82.9	84.7	85.8	87.2	(30)
(31)		5%	WB	79.8	79.9	80.7	80.2	78.7	76.7	74.9	74.7	75.6	77.7	79.4	80.0	(31)
(32)			MCD B	86.9	88.0	87.7	85.5	83.6	81.9	80.8	81.0	82.1	83.9	85.0	86.2	(32)
(33)		10%	WB	79.1	79.2	80.0	79.5	78.0	76.0	74.3	74.2	75.0	77.0	78.8	79.4	(33)
(34)			MCD B	85.9	86.9	86.4	84.5	82.5	81.1	80.1	80.3	81.5	83.1	84.4	85.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	14.0	13.0	9.9	9.3	10.3	11.1	13.0	14.6	14.2	12.7	12.5	(35)
(36)			MCDBR	14.3	15.4	15.4	11.7	10.6	11.0	12.0	13.8	15.6	15.8	14.2	13.6	(36)
(37)			MCWBR	4.5	5.3	5.5	5.0	5.0	4.7	4.7	5.1	5.7	5.8	5.9	5.2	(37)
(38)		5% WB	MCD BR	13.4	14.5	13.4	10.8	9.9	10.6	11.4	13.1	14.3	13.9	12.7	12.8	(38)
(39)			MCWBR	5.5	6.0	5.8	5.2	5.2	5.1	4.9	5.4	5.8	5.9	6.1	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.482	0.489	0.436	0.406	0.410	0.409	0.415	0.413	0.410	0.410	0.410	0.446	(40)	
(41)		taud	2.259	2.246	2.437	2.540	2.506	2.477	2.425	2.424	2.428	2.445	2.478	2.391	(41)	
(42)		Ebn at Noon	273	272	282	283	273	268	268	277	285	290	292	283	(42)	
(43)		Edh at Noon	46	47	38	33	33	33	35	36	38	38	37	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1906	1963	1824	1530	1428	1474	1512	1624	1810	1926	1932	1919	(44)	
(45)		RadStd	120	108	121	135	139	62	58	52	64	73	92	100	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page